

Total No. of Questions : 8]

SEAT No. :

PB-4751

[Total No. of Pages : 2

[6263]-435R

B.E. (Mechanical) (Honors)

**e-VEHICLE STANDARDS, CHARGING AND SAFETY
(2019 Pattern) (Semester - VIII) (302036MJ)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Describe Control architecture in EV charging. **[9]**
b) Describe EV integration in the power grid. What is the role of aggregator in EV? **[8]**

OR

- Q2)** a) Explain in detail Autonomous electric vehicles with advantages and applications. **[9]**
b) Explain with neat sketch Electric Vehicle Battery Charging. **[8]**

- Q3)** a) Explain Constant current discharge and static capacity test. **[9]**
b) State and explain various Testing procedures for EV batteries. **[9]**

OR

- Q4)** a) What are electrical and mechanical test procedures for safety requirements of traction batteries? **[9]**
b) Explain specific Requirements for L and M Category Electric Power Train Vehicles. **[9]**

P.T.O.

- Q5) a)** Explain the importance of thermal management systems in ensuring the safe operation of lithium-ion batteries in electric vehicles. **[9]**
- b)** Elaborate the concept of “Battery Pack and Module Safety”. **[8]**

OR

- Q6) a)** Write a note on : **[9]**
- i) Calorimetry
- ii) Electrochemical and Thermal Characterization of batteries
- b)** Explain the role of regulatory standards and industry guidelines in ensuring the safety. **[8]**

- Q7) a)** Differentiate between domestic and public charging infrastructure, highlighting their respective features and functionalities. **[9]**
- b)** Explain the concept of Battery Swapping with its types, advantages and disadvantages. **[9]**

OR

- Q8) a)** Explain the concept of multi-stakeholder governance in the context of EV charging infrastructure and its importance for effective management. **[9]**
- b)** Explain planning steps of location planning and land allocation for EV charging infrastructure. **[9]**

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